DEPARTMENT OF
COMPUTER SCIENCE

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PRIVACY-ENHANCED ONTOLOGIES

A PRACTICAL SSE M/M FRAMEWORK

MASTER IN COMPUTER SCIENCE

NOVA University Lisbon
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Abstract

1. What is the problem?
2. Why is this problem interesting/challenging?
3. What is the proposed approach/solution?
4. What results (implications/consequences) from the solution?

Keywords: Keyword 1, Keyword 2, Keyword 3, Keyword 4

Resumo

1. Qual é o problema?
2. Porque é que é um problema interessante/desafiante?
3. Qual é a proposta de abordagem/solução?
4. Quais são as consequências/resultados da solução proposta?

Palavras-chave: Palavra-chave 1, Palavra-chave 2, Palavra-chave 3, ...

Contents

Acronyms

SLA service level agreement 2

Symbols

1

Introduction

1.1 Context

In 2001, the term *Semantic Web* [3] was first used to define what would be the evolution of the *World Wide Web*. The traditional *Web* solved the challenge of distributing information at scale and globally, however it was designed for human use. Documents are easily accessible and readable by humans, but for automated agents little more than document retrieval is possible. Contrary to humans, machines cannot reason over the information they access, if they do not know the *semantics of the data*.

The *Semantic Web* addresses this issue, by redesigning the representation of information, in a way that not only preserves meaning and relationships but is also machine-readable. A concept is mapped to a unique identifier, and meaning is encoded in sets of triples, each triple consisting of a subject, predicate and object (similar to an elementary sentence). *Ontologies* provide a shared understanding of a domain by formalizing the specification of relevant concepts, and their relations, in such domain.

This novel method of knowledge representation differs from the previous, centralized way, and greatly simplifies the process of interconnecting different knowledge bases. Centralized representations tend to be more expensive to develop, usually needing to be built from scratch [12]. Furthermore, for sharing, both parties need to agree on the semantics. In contrast, using ontologies, information is semantically linked by default. In the *Semantic Web*, machines can truly communicate, share and reason about distinct knowledge bases.

Knowledge representation is essential in the industry [13] and ontologies help remove ambiguity in terminology. Nowadays, they are being developed and applied in a variety of domains [19] ranging from the health sector[1], to linguistics [2] and even being used by companies like NASA [2]. [3]

[1] NCI Thesaurus [16], developed by the National Cancer Institute (NCI) or the Gene Ontology (GO) [7], the world's largest source of information on the functions of genes, funded by the National Human Genome Research Institute.

[2] WordNet [9], a lexical database for English where related words and concepts are semantically linked.

[3] Observing the visual representation provided by the Linked Open Data Cloud [8] is a good exercise to perceive the dimension of adoption of this new paradigm.

To developed and maintain ontologies, advanced semantic editors are the standard tool for professionals, with Protégé [11] being a very popular example. These provide numerous functionalities, usually based on plugins (e.g. automatic verification of errors and consistency in the design; debugging tools; query answering; graphic visualization of data).

With ontologies gradually growing in dimension, the industry is currently evolving towards the use of collaborative solutions [18, 17] and outsourcing heavy computations, relying on cloud infrastructures or other shared computing platforms. These approaches, albeit inevitable, do present new security concerns and engineering challenges.

As of 2021, cloud adoption as risen more than what was previously forecasted, with 50 percent of all enterprise workloads hosted in *public clouds* and predictions of 7 percentage point gain in 2022 [15]. Research on secure cloud computing [4] studies such adoption but also discusses security concerns. Adoption can be explained with the cost efficiency[4], high scalability, flexibility and availability that cloud provides. Additionally, the pay-per-use model is an easy entry point to the ecosystem.

On the contrary, while delegation of responsibilities is beneficial for costs, it is prejudicial to control. Using cloud services implies trusting its owner. A service level agreement (SLA) defines, in accordance to metrics, the service levels of the system that the vendor should provide (e.g. percentage of availability), while also defining penalties in case of not achieving what was agreed-on. However, there still exist the risk of loss of privacy, potential privacy leaks and data breaches. This is especially concerning with sensitive data such as health records.

On the setting of outsourcing data to third parties we usually consider the *honest-but-curious*, or *passive*, adversary model. This adversary will follow the protocol properly but keep a record of all its intermediate computations[6]. Furthermore, it cannot do any other action except attempting to learn private information via observation of views of the protocol execution [5].

A first approach to secure data in the cloud would be to use *Encryption at Rest*, which naively encrypts all data stored in the untrusted servers. If a user wants to do to write or search operation over the data, they will have to download all the data, and decrypt it. This incurs in high-network requirements and is computationally expensive on the client-side, hindering all the benefits of using a cloud service to store the data. Nonetheless, this is the mechanism usually provided by database services [14, 1, 10].

[4]Cloud services reduce implementation and maintenance, power, cooling, floor space, storage and operational costs

1.2 Problem

1.3 Objectives

Briefly explain proposed solution. The scope of the solution, contributions, and how will it solve the problems mentioned in previous section.

1.4 Contributions

Enumerate the contributions.

1.5 Document Organization

Specify the document organization, per chapter.

2

Related Work

2.1 Semantic Web

2.1.1 Languages

2.1.1.1 RDF: Resource Description Framework

2.1.1.2 RDF Schema

2.1.1.3 OWL2: Web Ontology Language

2.1.2 Querying the Semantic Web

2.1.2.1 RDF Stores

2.1.2.2 SPARQL

2.1.2.3 SWRL

2.1.3 Tools & Development Software for Ontologies

2.1.3.1 Semantic Editors

2.1.3.2 Semantic Reasoners

2.2 Searchable Encryption

2.2.1 Security Definitions

2.2.2 Client/Server Model

2.2.3 Symmetric Searchable Encryption

2.2.4 Homomorphic Encryption

2.3 Summary

3

Approach to Elaboration

3.1 System Requirements

3.2 Privacy Guaranties

3.3 Architecture & System Model

3.3.1 Implementation Guidelines

3.3.2 Validation

4

Work Plan

Bibliography

[1] O. C. and/or its affiliates. *MySQL :: MySQL 8.0 reference manual :: 15.13 InnoDB data-at-rest encryption*. en. https://dev.mysql.com/doc/refman/8.0/en/innodb-data-encryption.html. Accessed: 2022-2-9 (cit. on p. 2).

[2] N. Ashish. "Semantic-Web Technology: Applications at NASA". In: *Dagstuhl Seminar Proceedings*. 2005 (cit. on p. 1).

[3] T. Berners-Lee, J. Hendler, and O. Lassila. *The semantic web*. Vol. 284. 2001. DOI: 10.1038/scientificamerican0501-34 (cit. on p. 1).

[4] M. Carroll, A. Van Der Merwe, and P. Kotzé. "Secure cloud computing: Benefits, risks and controls". In: *2011 Information Security for South Africa - Proceedings of the ISSA 2011 Conference*. 2011. DOI: 10.1109/ISSA.2011.6027519 (cit. on p. 2).

[5] D. Evans, V. Kolesnikov, and M. Rosulek. "A Pragmatic Introduction to Secure Multi-Party Computation". In: *Foundations and Trends® in Privacy and Security* 2.2-3 (2018), pp. 70–246. ISSN: 2474-1558. DOI: 10.1561/3300000019 (cit. on p. 2).

[6] O. Goldreich. *Foundations of Cryptography: Volume 2, Basic Applications*. en. Cambridge University Press, 2001. ISBN: 9780521830843 (cit. on p. 2).

[7] M. A. Harris et al. "The Gene Ontology (GO) database and informatics resource". en. In: *Nucleic Acids Res.* 32.Database issue (Jan. 2004), pp. D258–61. ISSN: 0305-1048, 1362-4962. DOI: 10.1093/nar/gkh036 (cit. on p. 1).

[8] J. P. McCrae. *The linked open data cloud*. en. https://lod-cloud.net/. Accessed: 2022-2-8 (cit. on p. 1).

[9] G. A. Miller. "WordNet: a lexical database for English". In: *Commun. ACM* 38.11 (Nov. 1995), pp. 39–41. ISSN: 0001-0782. DOI: 10.1145/219717.219748 (cit. on p. 1).

[10] MongoDB, Inc. *MongoDB Manual: Encryption at rest*. en. https://docs.mongodb.com/manual/core/security-encryption-at-rest/. Accessed: 2022-2-9 (cit. on p. 2).

[11] M. A. Musen and Protégé Team. "The Protégé Project: A Look Back and a Look Forward". en. In: *AI Matters* 1.4 (June 2015), pp. 4–12. ISSN: 2372-3483. DOI: 10.1145/2757001.2757003 (cit. on p. 2).

[12] R. Neches et al. "Enabling technology for knowledge sharing". In: *AI Magazine* 12.3 (1991). ISSN: 0738-4602 (cit. on p. 1).

[13] N. Noy et al. "Industry-scale knowledge graphs: Lessons and challenges". In: *Commun. ACM* 62.8 (2019). ISSN: 0001-0782, 1557-7317. DOI: 10.1145/3331166 (cit. on p. 1).

[14] Oracle Corporation and/or its affiliates. *MySQL enterprise encryption*. en. https://www.mysql.com/products/enterprise/encryption.html. Accessed: 2022-2-9 (cit. on p. 2).

[15] Security Compass. *The 2021 State of Cloud Adoption*. en. https://resources.securitycompass.com/reports/2021-state-of-cloud-adoption. Accessed: 2022-2-9. Sept. 2021 (cit. on p. 2).

[16] N. Sioutos et al. "NCI Thesaurus: a semantic model integrating cancer-related clinical and molecular information". en. In: *J. Biomed. Inform.* 40.1 (Feb. 2007), pp. 30–43. ISSN: 1532-0464, 1532-0480. DOI: 10.1016/j.jbi.2006.02.013 (cit. on p. 1).

[17] T. Tudorache, J. Vendetti, and N. F. Noy. "Web-Protégé: A lightweight OWL ontology editor for the web". In: *CEUR Workshop Proceedings*. Vol. 432. 2009 (cit. on p. 2).

[18] T. Tudorache et al. "Supporting collaborative ontology development in Protégé". In: *Lect. Notes Comput. Sci.* 5318 LNCS (2008), pp. 17–32. ISSN: 0302-9743, 1611-3349. DOI: 10.1007/978-3-540-88564-1_2 (cit. on p. 2).

[19] W3C. *Semantic Web Case Studies and Use Cases*. en. https://www.w3.org/2001/sw/sweo/public/UseCases/. Accessed: 2022-2-8 (cit. on p. 1).